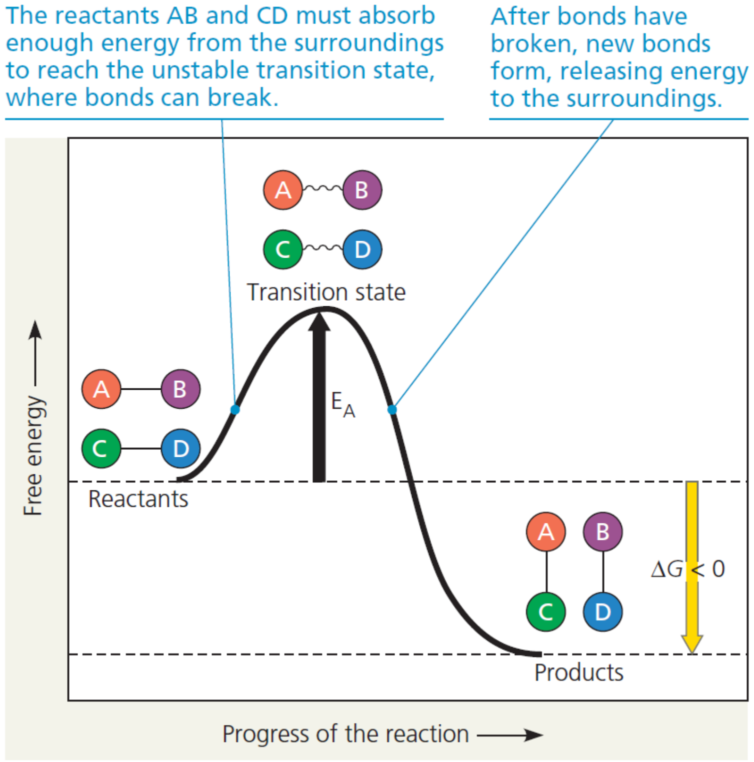
Enzymes

- macromolecule that acts as a **catalyst**, a chemical agent that speeds up a reaction without being consumed by the reaction.
- They do not add any energy to reactions (hence cannot make an endergonic reaction work!)

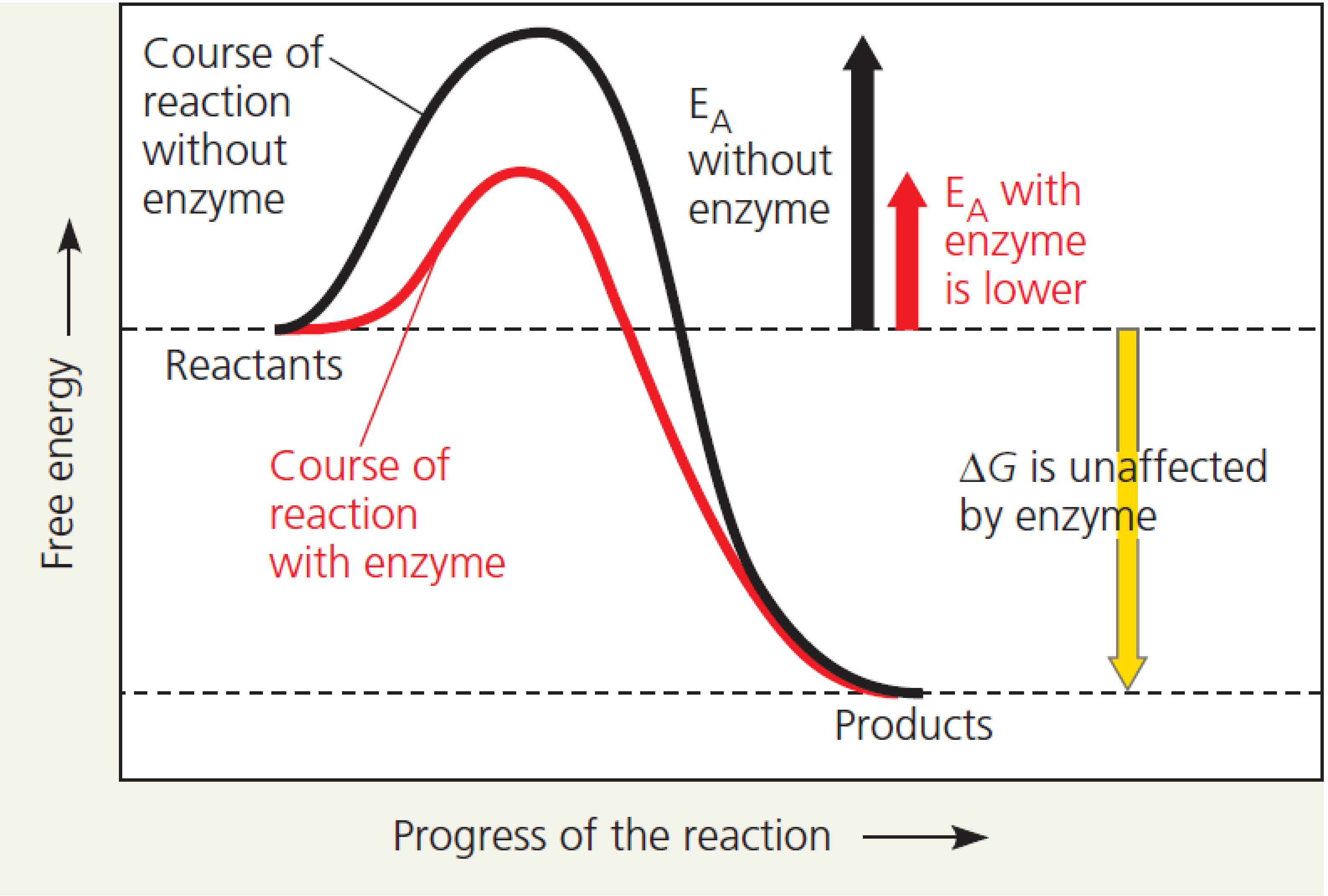
Figure 8.13 Energy profile of an exergonic reaction. The "molecules" are hypothetical, with A, B, C, and D representing portions of the molecules. Thermodynamically, this is an exergonic reaction, with a negative ΔG, and the reaction occurs spontaneously. However, the activation energy (E_A) provides a barrier that determines the rate of the reaction.



$$AB + CD \rightarrow AC + BD$$

Reactants   Products

Figure 8.14 The effect of an enzyme on activation energy. Without affecting the free-energy change (ΔG) for a reaction, an enzyme speeds the reaction by reducing its activation energy (E_A).



an enzyme cannot change the G for a reaction; it cannot make an endergonic reaction exergonic
would eventually occur anyway, but this enables the cell
to have a dynamic metabolism, routing chemicals smoothly through metabolic pathways. Also, enzymes are very specific
for the reactions they catalyze, so they determine which chemical processes will be going on in the cell at any given time.

Substrate Specificity of Enzymes

Enzyme + Substrate(s) $\rightleftharpoons$ Enzyme-substrate complex $\rightleftharpoons$ Enzyme + Product(s)

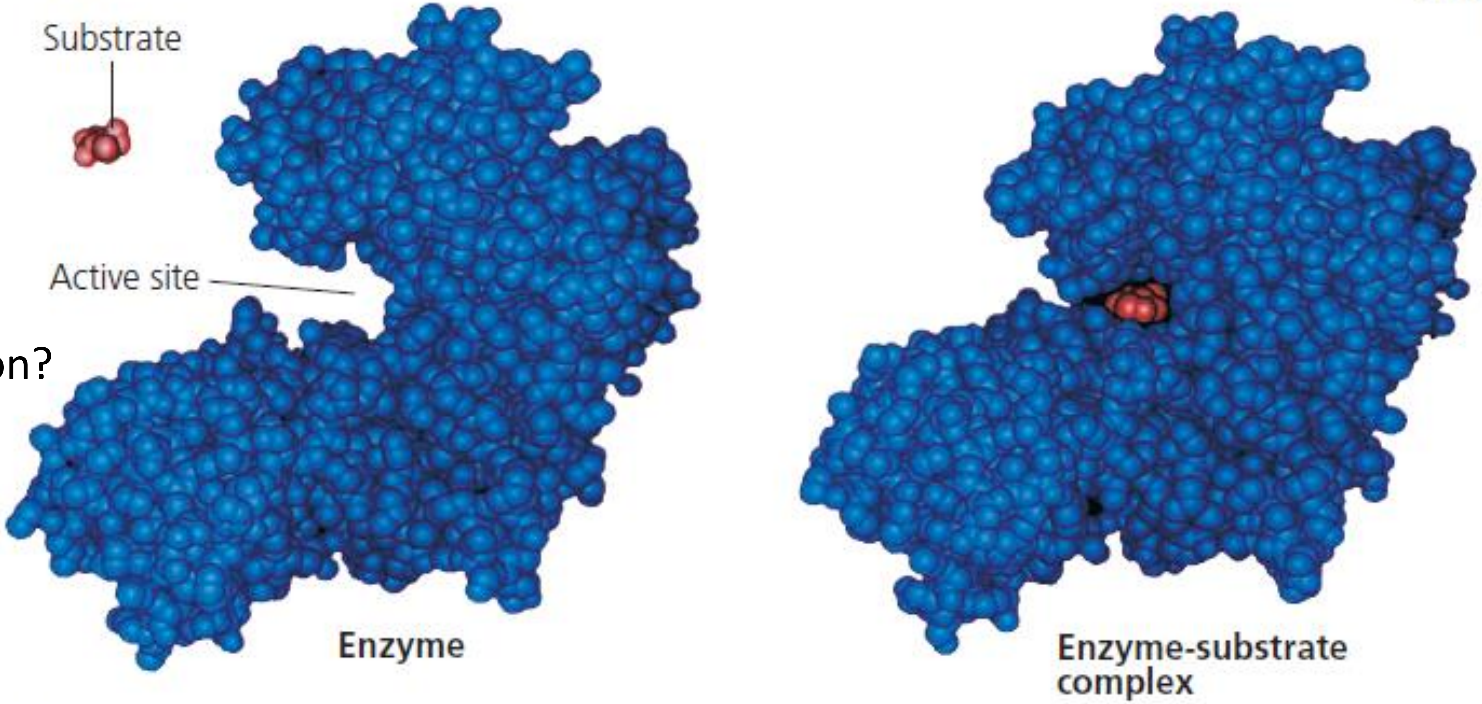
each enzyme is very specific

What accounts for this molecular recognition?

The specificity of an enzyme is attributed to a complementary fit between the shape of its active site and the shape of the substrate.

Figure 8.15 Induced fit between an enzyme and its substrate.

shape. In fact,
that en
"da



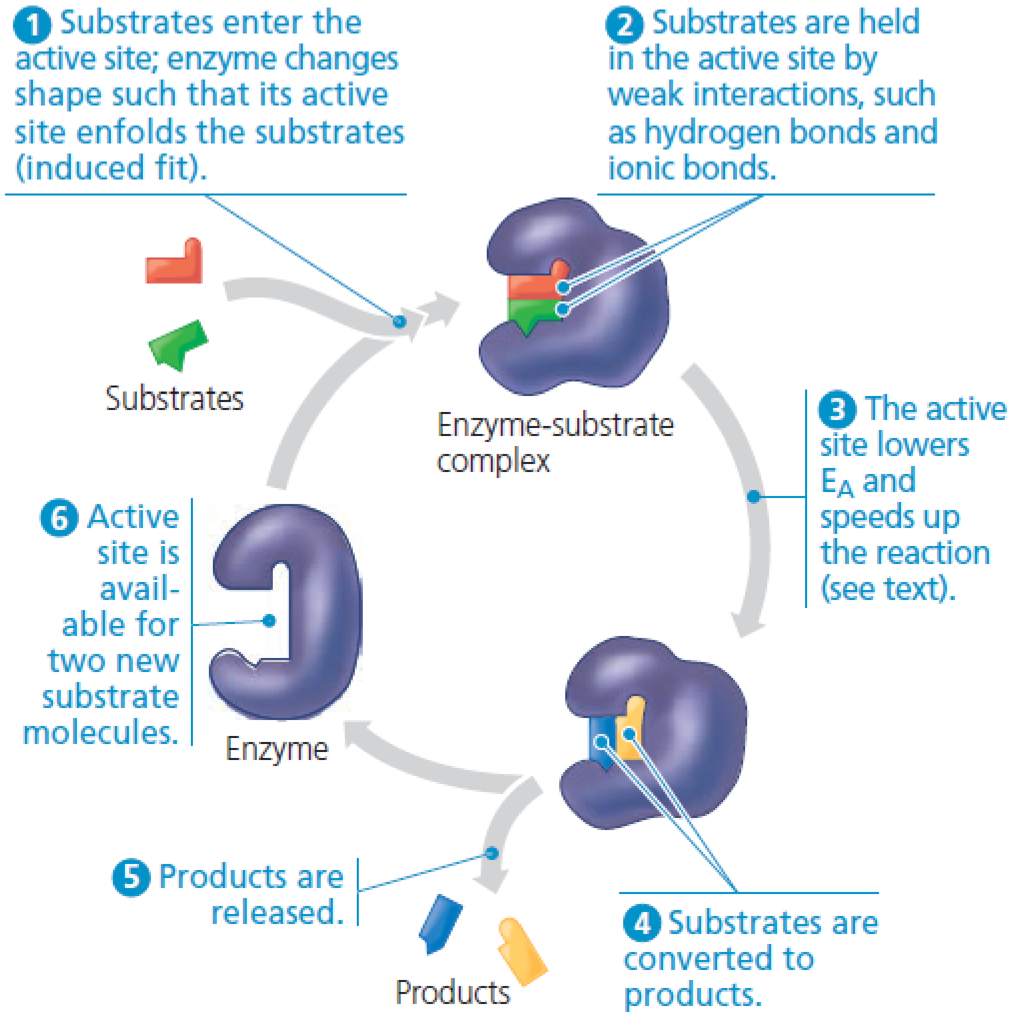
(a) In this space-filling model of the enzyme hexokinase (blue), the active site forms a groove on the surface. The enzyme's substrate is glucose (red).

(b) When the substrate enters the active site, it forms weak bonds with the enzyme, inducing a change in the shape of the protein. This change allows additional weak bonds to form, causing the active site to enfold the substrate and hold it in place.

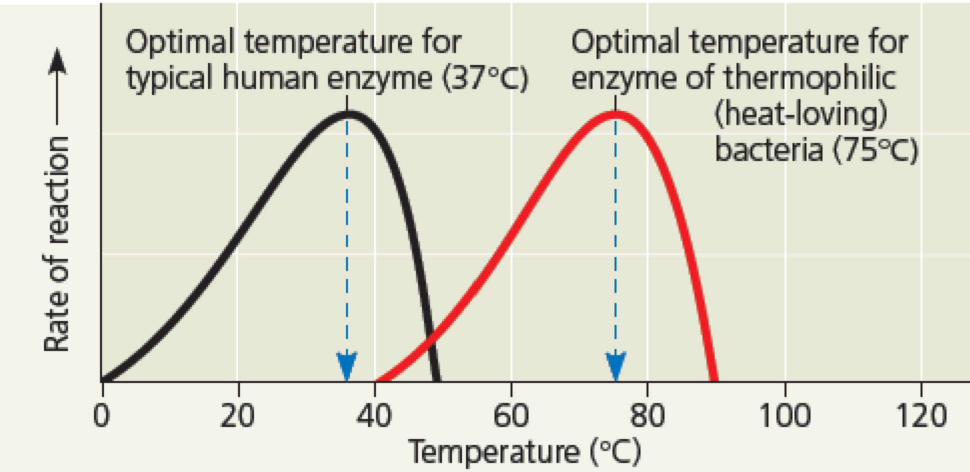
- The induced-fit model is an adjustment to the lock-and-key model and explains how enzymes and substrates undergo dynamic modifications during the transition state to increase the affinity of the substrate for the active site.

Catalysis in the Enzyme’s Active Site

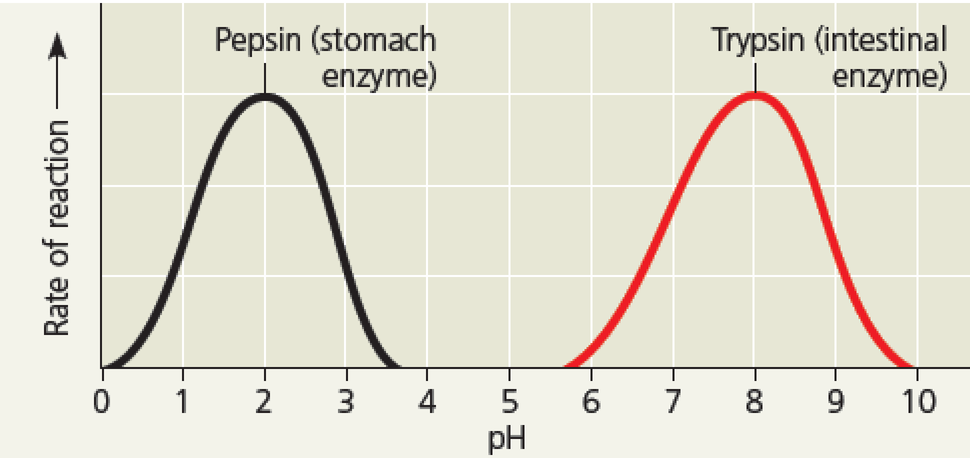
Figure 8.16 The active site and catalytic cycle of an enzyme. An enzyme can convert one or more reactant molecules to one or more product molecules. The enzyme shown here converts two substrate molecules to two product molecules.



Factors affecting enzyme activity



(a) The photo shows thermophilic cyanobacteria (green) thriving in the hot water of a Nevada geyser. The graph compares the optimal temperatures for an enzyme from the thermophilic bacterium *Thermus oshimai* (75°C) and human enzymes (body temperature, 37°C).



(b) This graph shows the rate of reaction for two digestive enzymes over a range of pH values.

Enzyme regulation

- Or controlling a chemical reaction through changing the enzyme activity (!)
    - Enzyme activation – by favoring the binding of substrate to the active site
    - Enzyme inactivation (inhibition) – by blocking the substrate from binding to active site

- Help treating diseases by developing drugs that can change the enzyme activity
    - Very efficient since substrates and enzymes are HIGHLY specific

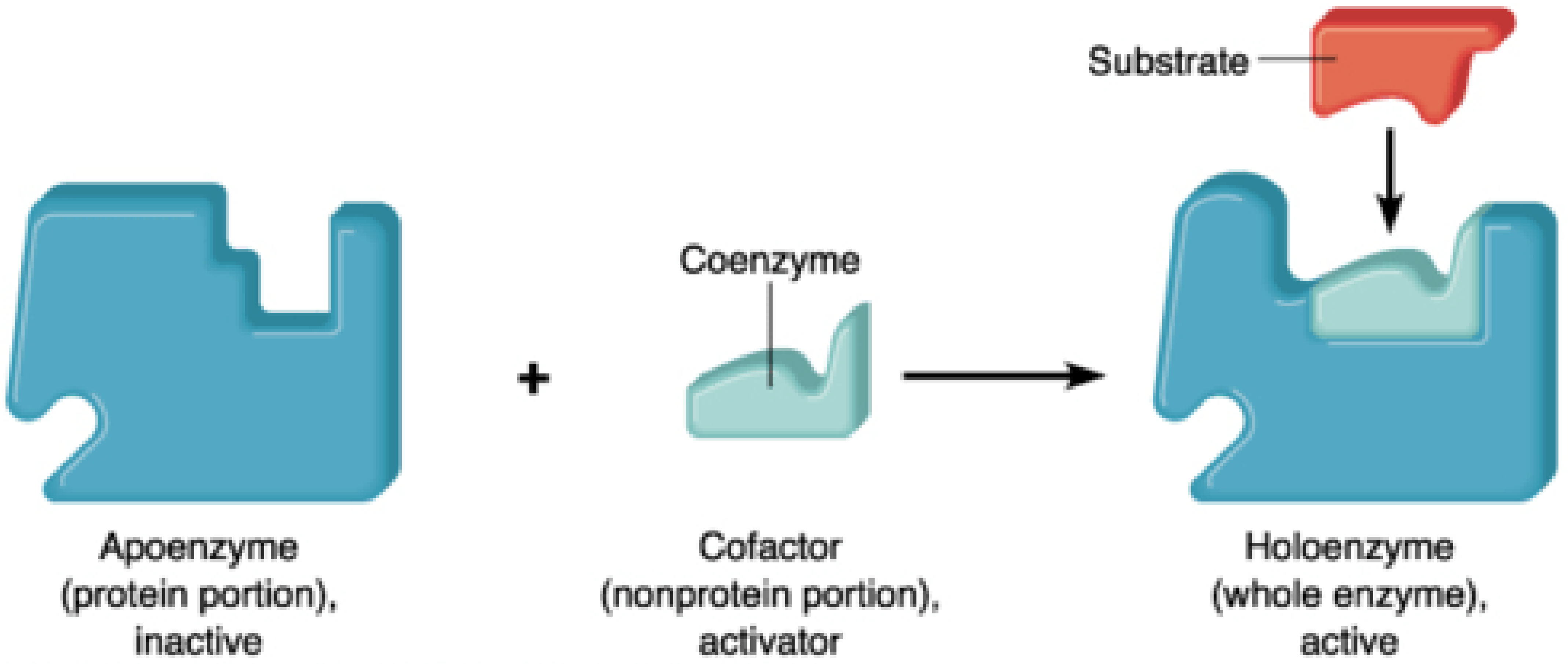
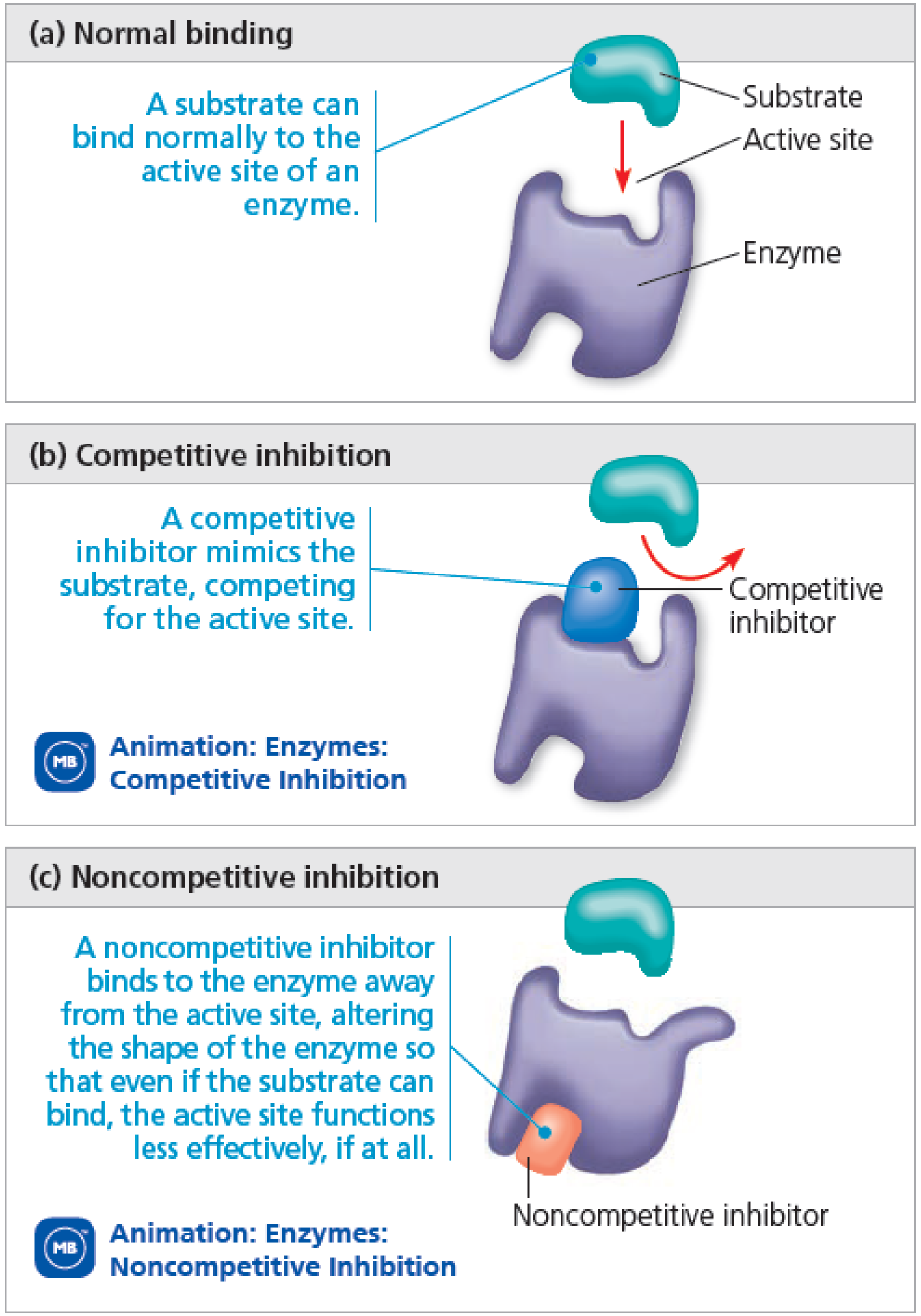
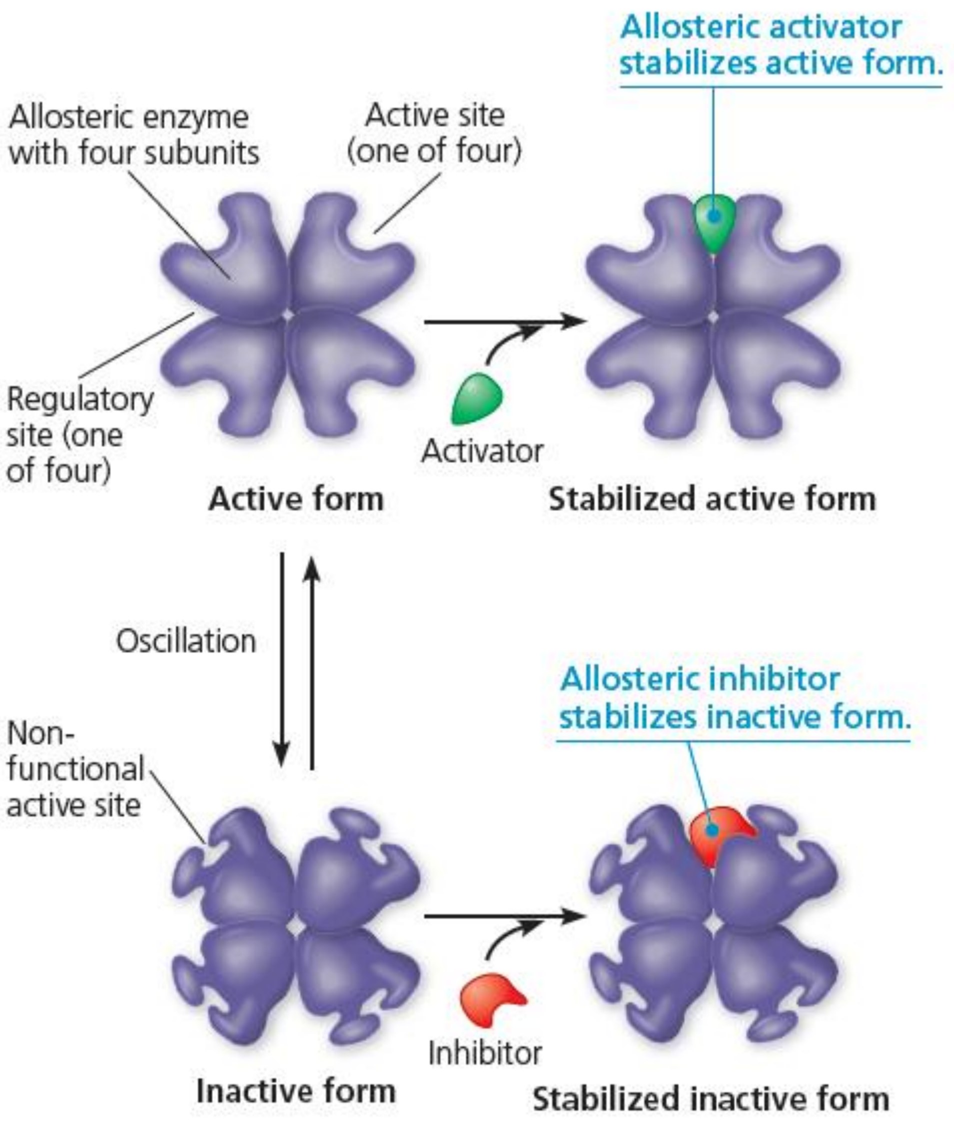

Substrate
Coenzyme
Apoenzyme
(protein portion),
inactive
Cofactor
(nonprotein portion),
activator
Holoenzyme
(whole enzyme),
active

Figure 8.18 Inhibition of enzyme activity.

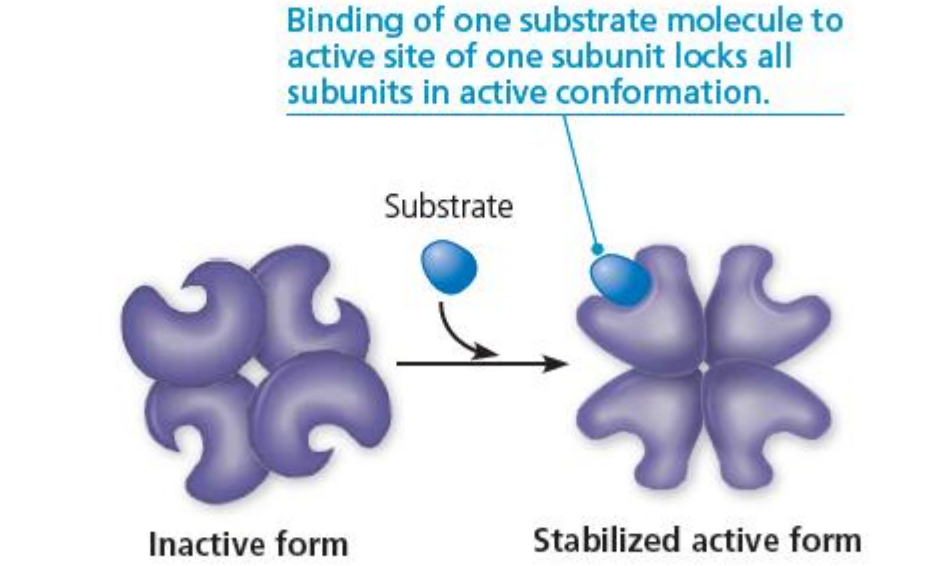


(a) Allosteric activators and inhibitors



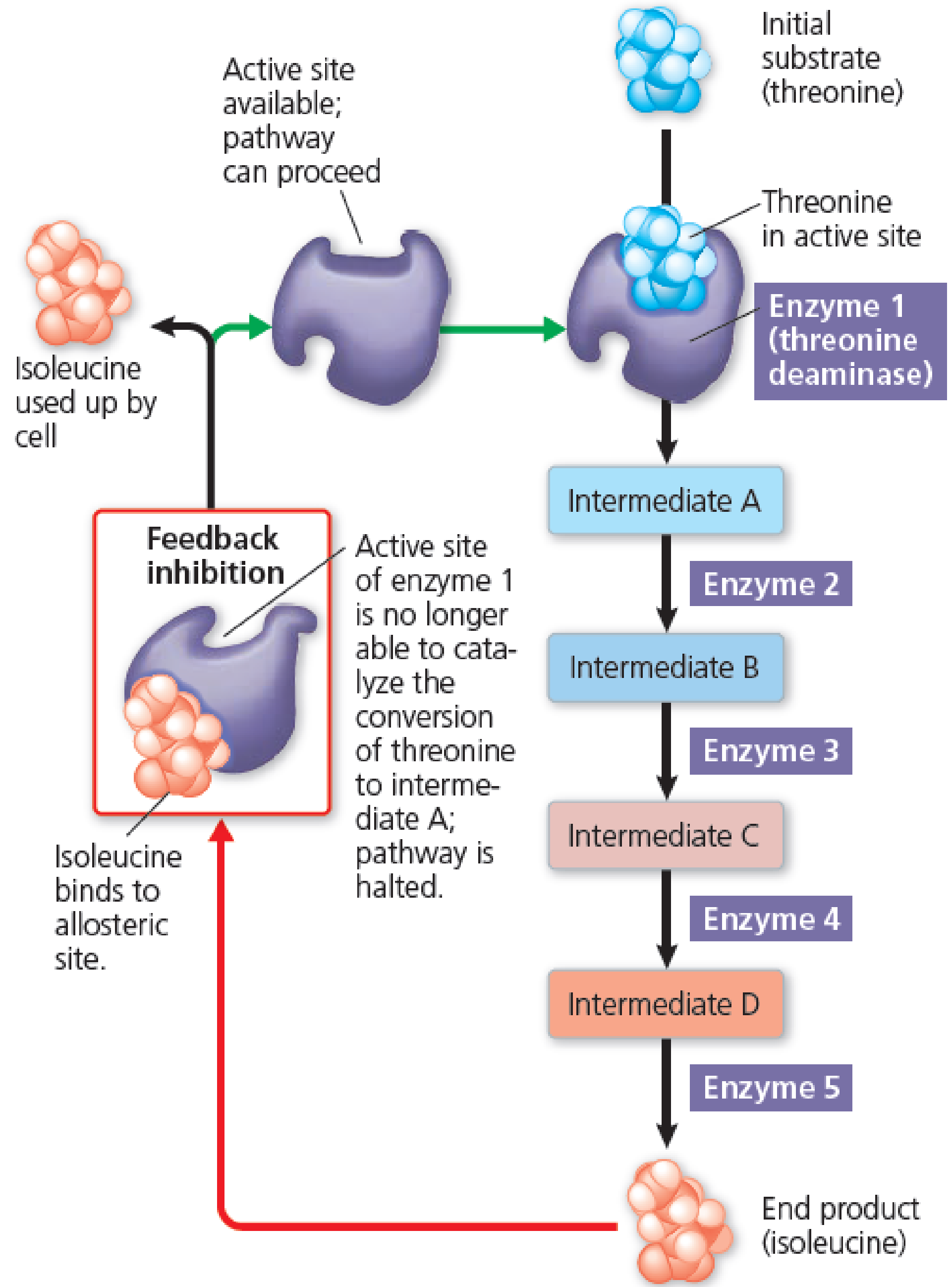
At low concentrations, activators and inhibitors dissociate from the enzyme. The enzyme can then oscillate again.

(b) Cooperativity: another type of allosteric activation



The inactive form shown on the left oscillates with the active form when the active form is not stabilized by substrate.

Figure 8.21 Feedback inhibition in isoleucine synthesis.



Energy flow and chemical recycling in ecosystems.